Ting Yu Tsai

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Education

- **University at Albany - SUNY** | **USA** | Ph.D. in Computer Science. Aug. 2023 ~ Present
- **National Taiwan University** | **Taiwan** | M.S in Biomedical Electronics and Bioinformatics. Aug. 2019 ~ Jul. 2021

Work Experience

- **Research Assistant** | *University at Albany-SUNY* Jan. 2024~Dec. 2024
  - **Project:** Cardiovascular Image Segmentation | **Language:** Python | **Library:** Pytorch
  - Built and evaluated deep learning models for cardiovascular image segmentation, improving model robustness through SAM and uncertainty-aware loss design.
- **Process Engineer** | *Taiwan Semiconductor Manufacturing Co., Ltd. (TSMC)* Nov. 2021~Jun. 2022
  - Supported semiconductor manufacturing operations by testing equipment, optimizing system processes, and performing risk assessments to improve efficiency and reliability.
- **Software Engineer Intern** | *Test Research Inc.* Aug. 2019~Jul. 2021
  - **Project:** Super Resolution for X-Ray Images | **Language:** Python | **Library:** TensorFlow, Tensorlayer
  - Developed an X-ray image super-resolution model that outperformed five classic AI models and four interpolation methods in image quality and processing speed.
- **Software Engineer Intern** | *Chung Shan Medical University Hospital* Jul. 2018~Aug. 2018
  - **Project:** Automatic Head and Neck CT image Organ Contour Selection System | **Language:** Python
  - Created an automated head and neck CT organ contouring system that improved radiation oncology workflow efficiency by reducing selection time by 70%.

Projects

- **IntelliCardiac** | *2025*
  - Developed an end-to-end clinical-style cardiac MRI platform for automated three cardiac components segmentation and five classes disease classification, with real-time visualization and workflow-ready deployment.
- **An Uncertainty-Aware Generalization Framework for Cardiovascular Image Segmentation** | *2024*
  - Developed and benchmarked UU-Mamba for cardiovascular segmentation, integrating Sharpness-Aware Minimization (SAM) with an uncertainty-aware composite loss, achieving SOTA performance on ACDC, ImageCAS, and Aorta.
- **Automatic Image Exposure Classification and Improvement System with Deep Learning Algorithms** | *2021*
  - The accuracy of the classification function in the system is 0.91. The enhanced image quality is sure to compete with those taken by cameras.

Publications

- Yu, A., **Tsai, T.Y.**, Zhang, Z., Lu, W., Ye, F.X.-F., & Chang, M. (2026). ReDiPrune: Relevance-Diversity Pre-Projection Token Pruning for Efficient Multimodal LLMs. Under review for the European Conference on Computer Vision (ECCV).
- **Tsai, T.Y.**, Yu, A., Lee, L., Ye, F.X.-F., Shin, D.S., Kao, T.J., Li, X., & Chang, M. (2026). RatSeizure: A Benchmark and Saliency-Context Transformer for Rat Seizure Localization. Under review for the International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI).
- **Tsai, T.Y.**, Yu, A., Maadugundu, M.S., Mohima, I.J., Barsha, U.H., Chen, M.F., Prabhakaran, B., & Chang, M. (2025). IntelliCardiac: An Intelligent Platform for Cardiac Image Segmentation and Classification. 2025 IEEE 8th International Conference on Multimedia Information Processing and Retrieval (MIPR), 109-115.
- Wang, X., **Tsai, T.Y.,** Lin, L., Guo, H., Hu, S., Chang, M., Atrey, P.K., & Lyu, S. (2025). Spotting the Fakes: A Deep Dive into GAN-Generated Face Detection. ACM Transactions on Multimedia Computing, Communications and Applications, 21, 1-24.
- **Tsai, T.Y.**, Lin, L., Hu, S., Chang, M., Zhu, H., & Wang, X. (2024). UU-Mamba: Uncertainty-aware U-Mamba for Cardiac Image Segmentation. 2024 IEEE 7th International Conference on Multimedia Information Processing and Retrieval (MIPR), 267-273.